

(12) **United States Patent**  
**Kozinski et al.**

(10) **Patent No.: US 12,398,950 B2**  
(45) **Date of Patent: Aug. 26, 2025**

(54) **METHOD OF OPERATING A LIGHTING ASSEMBLY IN A REFRIGERATOR APPLIANCE**

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(\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 182 days.

(21) Appl. No.: **18/357,396**

(22) Filed: **Jul. 24, 2023**

(65) **Prior Publication Data**

US 2025/0035372 A1 Jan. 30, 2025

(51) **Int. Cl.**  
**F25D 27/00** (2006.01)  
**H04N 23/74** (2023.01)  
**H05B 47/12** (2020.01)  
**H05B 47/155** (2020.01)  
**H05B 47/175** (2020.01)

(52) **U.S. Cl.**  
CPC ..... **F25D 27/005** (2013.01); **H04N 23/74**  
(2023.01); **H05B 47/12** (2020.01); **H05B**  
**47/155** (2020.01); **H05B 47/175** (2020.01);  
**F25D 2500/06** (2013.01); **F25D 2700/06**  
(2013.01)

(58) **Field of Classification Search**  
CPC ..... **F25D 27/00**; **F25D 27/005**; **F25D 29/008**;  
**F25D 2327/00**; **F25D 2500/06**; **F25D**  
**2700/06**

See application file for complete search history.

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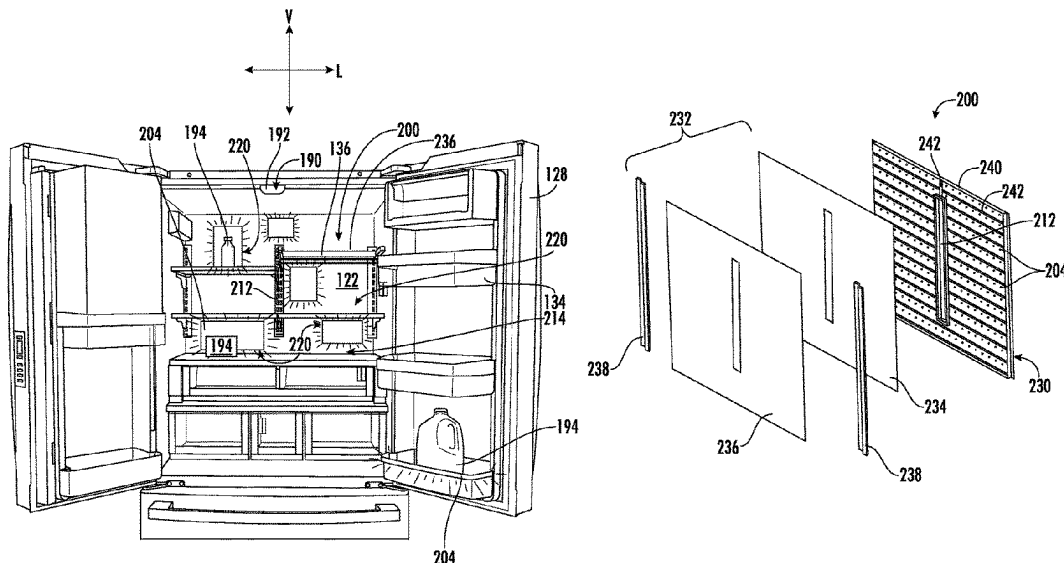
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(57) **ABSTRACT**

A refrigerator appliance includes a chilled chamber comprising a plurality of storage locations, a lighting assembly comprising a plurality of lighting zones corresponding to the plurality of storage locations, and a controller in operative communication with the lighting assembly. The controller is configured to receive a request to identify an object stored in the chilled chamber, determine that the object is located within a target location of the plurality of storage locations, identify a target lighting zone of the plurality of lighting zones that corresponds to the target location, and illuminate the target lighting zone.

**20 Claims, 6 Drawing Sheets**



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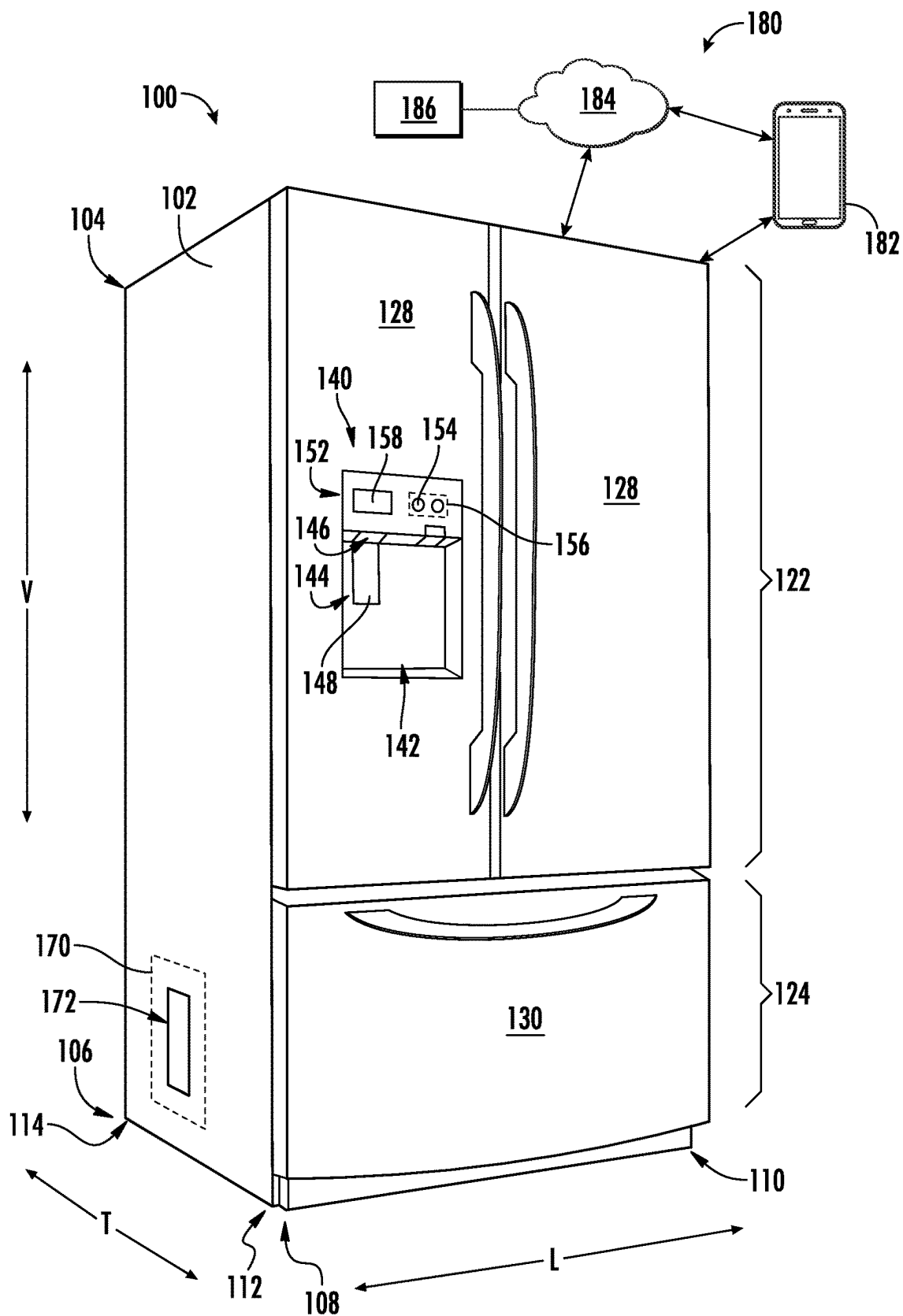


FIG. 1

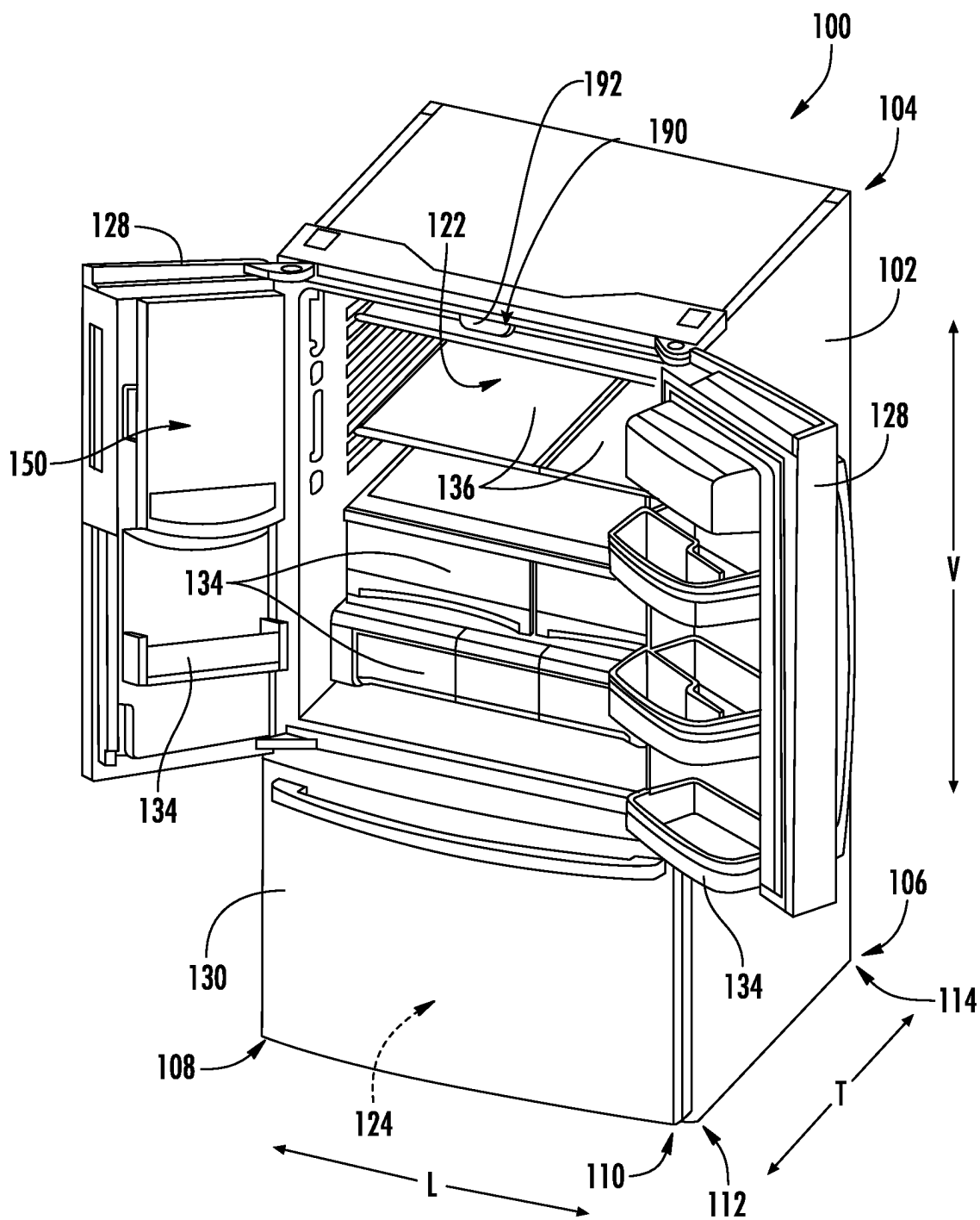


FIG. 2

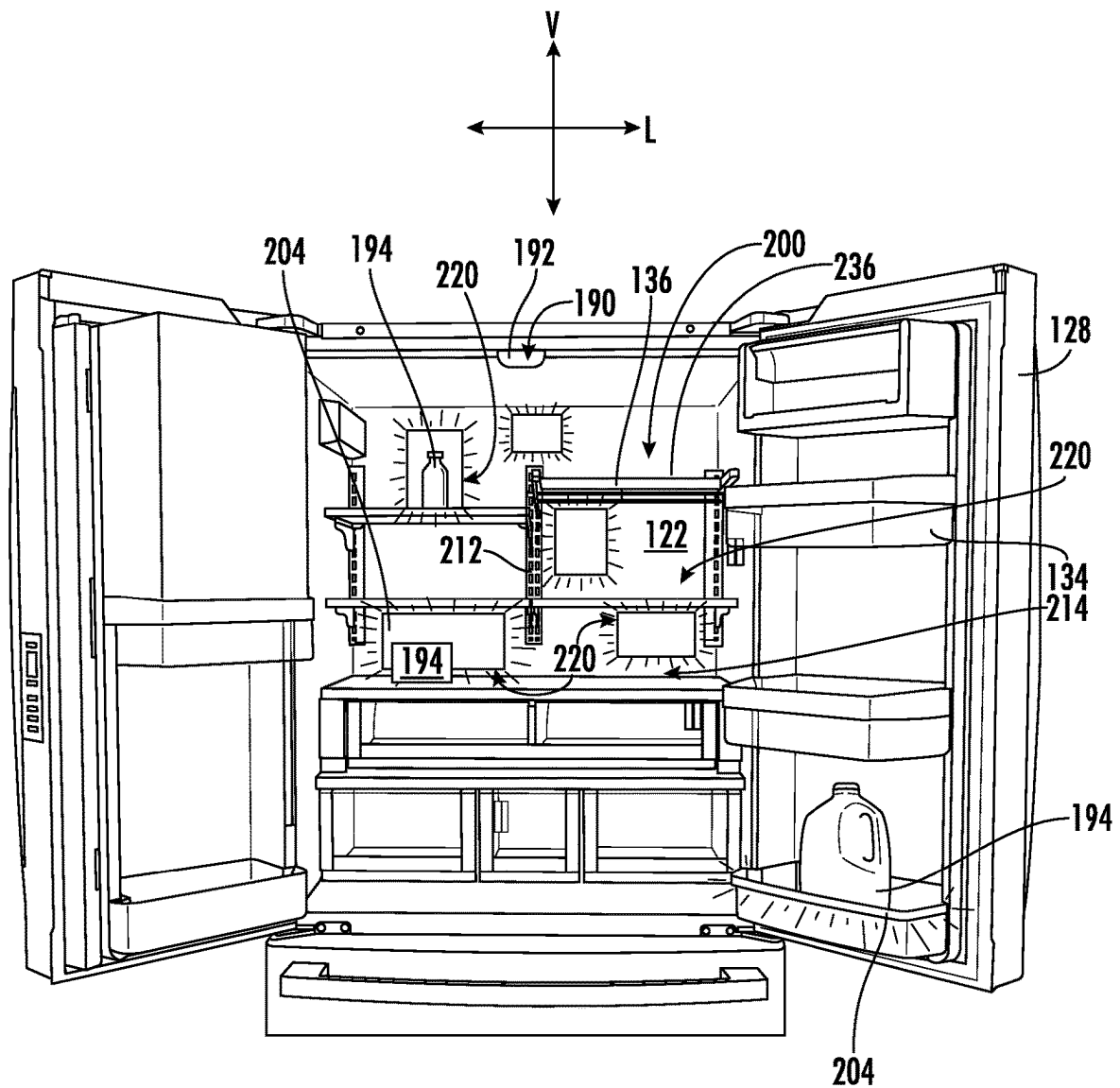


FIG. 3

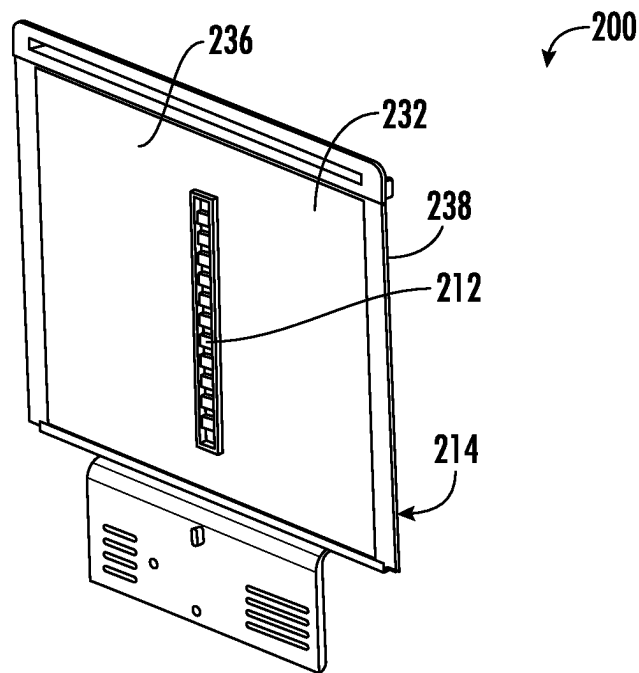


FIG. 4

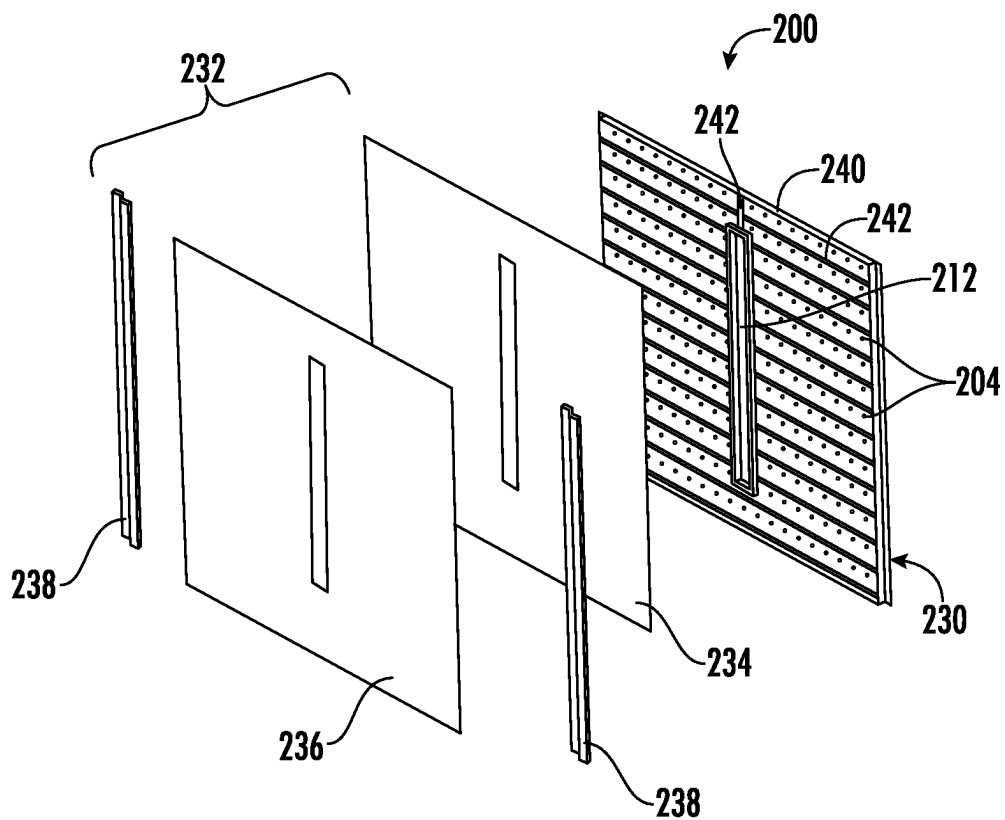


FIG. 5

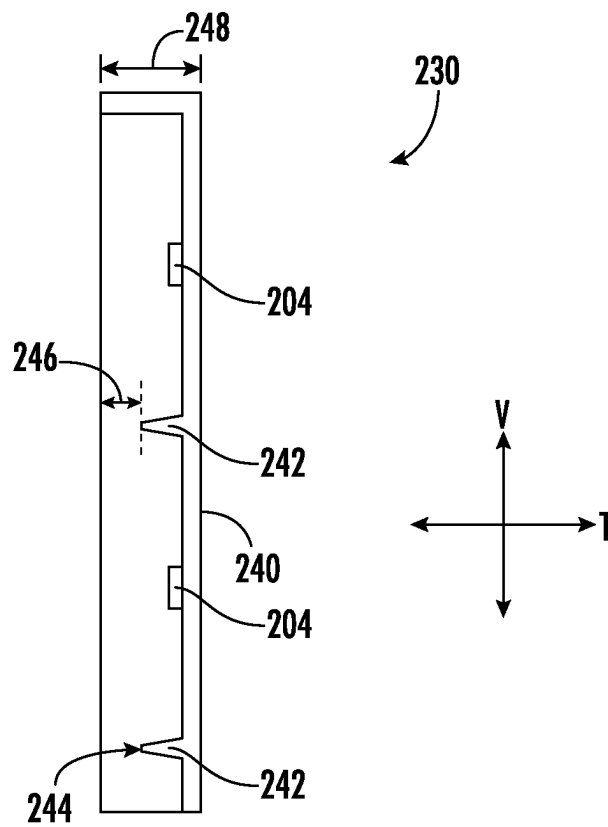
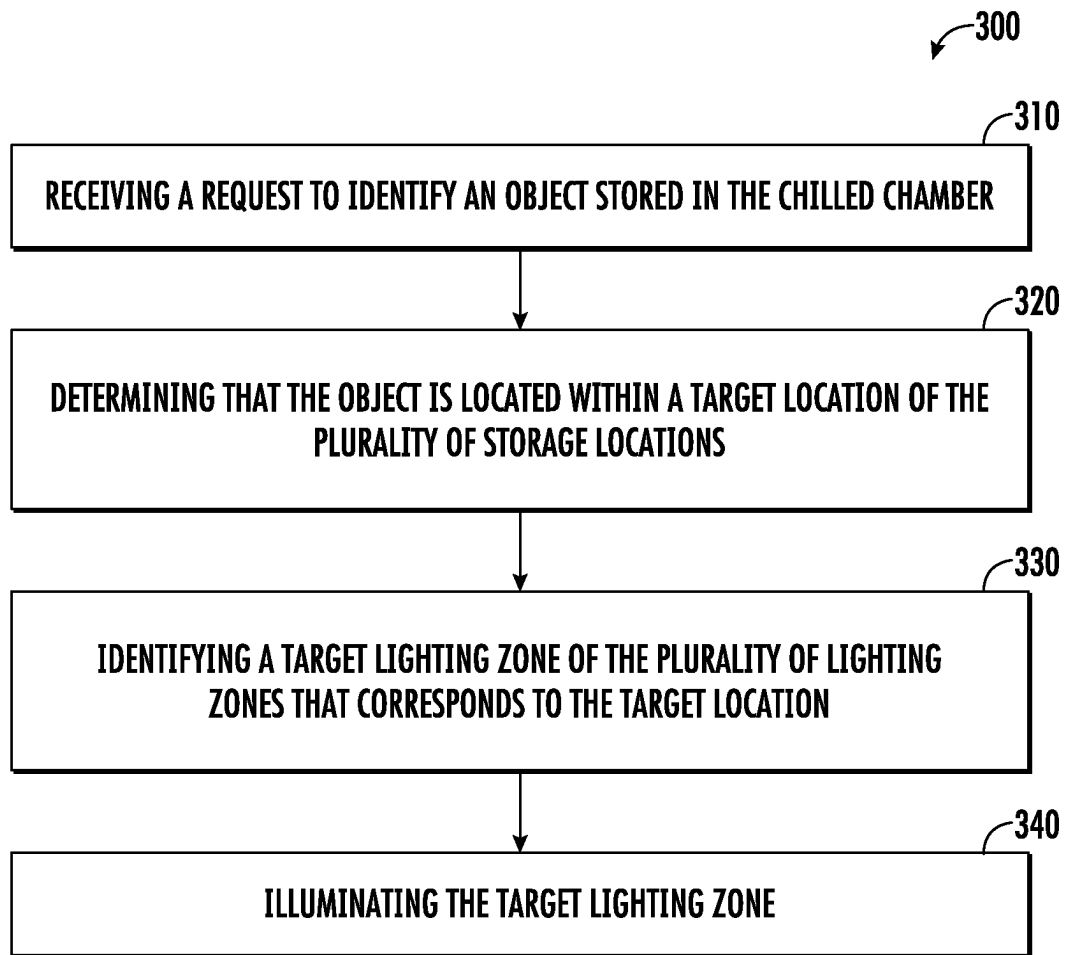


FIG. 6

**FIG. 7**



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## METHOD OF OPERATING A LIGHTING ASSEMBLY IN A REFRIGERATOR APPLIANCE

### FIELD OF THE INVENTION

The present subject matter relates generally to refrigerator appliances, and more particularly to lighting systems for refrigerator appliances.

### BACKGROUND OF THE INVENTION

Refrigerator appliances generally include a cabinet that defines a chilled chamber for receipt of food articles for storage. In addition, refrigerator appliances include one or more doors rotatably hinged to the cabinet to permit selective access to food items stored in chilled chamber(s). The refrigerator appliances can also include various storage components mounted within the chilled chamber and designed to facilitate storage of food items therein. Such storage components can include racks, bins, shelves, or drawers that receive food items and assist with organizing and arranging of such food items within the chilled chamber.

In addition, conventional refrigerator appliances include lighting systems that illuminate the chilled chamber. However, these conventional lighting systems are intended only to improve visibility within the chamber. In this regard, these conventional lighting systems are passively operated, e.g., the lighting system is activated when a door switch indicates that the door is open and the lighting system is deactivated when the door switch indicates that the door is closed. Moreover, conventional lighting systems lack versatility and the ability to communicate information regarding appliance operation. In this regard, even when the lighting systems are energized, they operate at a single, uniform intensity and color throughout the chilled chamber.

Accordingly, a refrigerator appliance with an improved lighting system would be useful. More particularly, a lighting system for a refrigerator appliance that provides versatile lighting configurations, improved aesthetics, and a more informative user experience would be particularly beneficial.

### BRIEF DESCRIPTION OF THE INVENTION

Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

In one exemplary embodiment, a refrigerator appliance is provided including a cabinet, a chilled chamber defined within the cabinet, the chilled chamber comprising a plurality of storage locations, a lighting assembly comprising a plurality of lighting zones corresponding to the plurality of storage locations, and a controller in operative communication with the lighting assembly. The controller is configured to receive a request to identify an object stored in the chilled chamber, determine that the object is located within a target location of the plurality of storage locations, identify a target lighting zone of the plurality of lighting zones that corresponds to the target location, and illuminate the target lighting zone.

In another exemplary embodiment, a method of operating a lighting assembly of a refrigerator appliance is provided. The refrigerator appliance includes a chilled chamber defined within a cabinet, the chilled chamber comprising a plurality of storage locations, and the lighting assembly

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comprising a plurality of lighting zones corresponding to the plurality of storage locations. The method includes receiving a request to identify an object stored in the chilled chamber, determining that the object is located within a target location of the plurality of storage locations, identifying a target lighting zone of the plurality of lighting zones that corresponds to the target location, and illuminating the target lighting zone.

These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

### BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. 1 provides a perspective view of a refrigerator appliance according to an exemplary embodiment of the present subject matter.

FIG. 2 provides a perspective view of the exemplary refrigerator appliance of FIG. 1, with the doors of the fresh food chamber shown in an open position.

FIG. 3 provides a front view of the exemplary refrigerator appliance of FIG. 1, with the doors of the fresh food chamber shown in an open position.

FIG. 4 provides a perspective view of a lighting assembly of the exemplary refrigerator appliance of FIG. 1 according to an exemplary embodiment of the present subject matter.

FIG. 5 provides an exploded view of the exemplary lighting assembly of FIG. 4 according to an exemplary embodiment of the present subject matter.

FIG. 6 provides a cross-sectional view of a backlight frame of the exemplary lighting assembly of FIG. 5 according to an exemplary embodiment of the present subject matter.

FIG. 7 provides a method of operating a refrigerator appliance and a lighting assembly according to an exemplary embodiment of the present subject matter.

Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

### DETAILED DESCRIPTION

Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

As used herein, the terms "first," "second," and "third" may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms

“upstream” and “downstream” refer to the relative flow direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the flow direction from which the fluid flows, and “downstream” refers to the flow direction to which the fluid flows. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”).

Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. For example, the approximating language may refer to being within a 10 percent margin.

FIG. 1 provides a perspective view of a refrigerator appliance 100 according to an exemplary embodiment of the present subject matter. Refrigerator appliance 100 includes a housing or cabinet 102 that extends between a top 104 and a bottom 106 along a vertical direction V, between a first side 108 and a second side 110 along a lateral direction L, and between a front side 112 and a rear side 114 along a transverse direction T. Each of the vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular to one another and form an orthogonal direction system.

Cabinet 102 defines chilled chambers for receipt of food items for storage. In particular, cabinet 102 defines fresh food chamber 122 positioned at or adjacent top 104 of cabinet 102 and a freezer chamber 124 arranged at or adjacent bottom 106 of cabinet 102. As such, refrigerator appliance 100 is generally referred to as a bottom mount refrigerator. It is recognized, however, that the benefits of the present disclosure apply to other types and styles of refrigerator appliances such as, e.g., a top mount refrigerator appliance, a side-by-side style refrigerator appliance, or a single door refrigerator appliance. Moreover, aspects of the present subject matter may be applied to other appliances as well, such as other appliances including fluid dispensers. Consequently, the description set forth herein is for illustrative purposes only and is not intended to be limiting in any aspect to any particular appliance or configuration.

Refrigerator doors 128 are rotatably hinged to an edge of cabinet 102 for selectively accessing fresh food chamber 122. In addition, a freezer door 130 is arranged below refrigerator doors 128 for selectively accessing freezer chamber 124. Freezer door 130 is coupled to a freezer drawer (not shown) slidably mounted within freezer chamber 124. To prevent leakage of cool air, refrigerator doors 128, freezer door 130, and/or cabinet 102 may define one or more sealing mechanisms (e.g., rubber gaskets, not shown) at the interface where the doors 128, 130 meet cabinet 102. It should be appreciated that doors having a different style, position, or configuration are possible within the scope of the present subject matter.

FIG. 2 provides a perspective view of refrigerator appliance 100 shown with refrigerator doors 128 in the open position. As shown in FIG. 2, various storage components are mounted within fresh food chamber 122 to facilitate storage of food items therein as will be understood by those skilled in the art. In particular, the storage components may include bins 134 and shelves 136. Each of these storage

components are configured for receipt of food items (e.g., beverages and/or solid food items) and may assist with organizing such food items. As illustrated, bins 134 may be mounted on refrigerator doors 128 or may slide into a receiving space in fresh food chamber 122. It should be appreciated that the illustrated storage components are used only for the purpose of explanation and that other storage components may be used and may have different sizes, shapes, and configurations.

Referring again to FIG. 1, a dispensing assembly 140 will be described according to exemplary embodiments of the present subject matter. Although several different exemplary embodiments of dispensing assembly 140 will be illustrated and described, similar reference numerals may be used to refer to similar components and features. Dispensing assembly 140 is generally configured for dispensing liquid water and/or ice. Although an exemplary dispensing assembly 140 is illustrated and described herein, it should be appreciated that variations and modifications may be made to dispensing assembly 140 while remaining within the present subject matter.

Dispensing assembly 140 and its various components may be positioned at least in part within a dispenser recess 142 defined on one of refrigerator doors 128. In this regard, dispenser recess 142 is defined on a front side 112 of refrigerator appliance 100 such that a user may operate dispensing assembly 140 without opening refrigerator door 128. In addition, dispenser recess 142 is positioned at a predetermined elevation convenient for a user to access ice and enabling the user to access ice without the need to bend-over. In the exemplary embodiment, dispenser recess 142 is positioned at a level that approximates the chest level of a user.

Dispensing assembly 140 includes an ice dispenser 144 including a discharging outlet 146 for discharging ice from dispensing assembly 140. An actuating mechanism 148, shown as a paddle, is mounted below discharging outlet 146 for operating ice or water dispenser 144. In alternative exemplary embodiments, any suitable actuating mechanism may be used to operate ice dispenser 144. For example, ice dispenser 144 can include a sensor (such as an ultrasonic sensor) or a button rather than the paddle. Discharging outlet 146 and actuating mechanism 148 are an external part of ice dispenser 144 and are mounted in dispenser recess 142. By contrast, refrigerator door 128 may define an icebox compartment 150 (FIG. 2) housing an icemaker and an ice storage bin (not shown) that are configured to supply ice to dispenser recess 142.

A control panel 152 is provided for controlling the mode of operation. For example, control panel 152 includes one or more selector inputs 154, such as knobs, buttons, touch-screen interfaces, etc., such as a water dispensing button and an ice-dispensing button, for selecting a desired mode of operation such as crushed or non-crushed ice. In addition, inputs 154 may be used to specify a fill volume or method of operating dispensing assembly 140. In this regard, inputs 154 may be in communication with a processing device or controller 156. Signals generated in controller 156 operate refrigerator appliance 100 and dispensing assembly 140 in response to selector inputs 154. Additionally, a display 158, such as an indicator light or a screen, may be provided on control panel 152. Display 158 may be in communication with controller 156, and may display information in response to signals from controller 156.

As used herein, “processing device” or “controller” may refer to one or more microprocessors or semiconductor devices and is not restricted necessarily to a single element.

The processing device can be programmed to operate refrigerator appliance **100**, dispensing assembly **140** and other components of refrigerator appliance **100**. The processing device may include, or be associated with, one or more memory elements (e.g., non-transitory storage media). In some such embodiments, the memory elements include electrically erasable, programmable read only memory (EEPROM). Generally, the memory elements can store information accessible processing device, including instructions that can be executed by processing device. Optionally, the instructions can be software or any set of instructions and/or data that when executed by the processing device, cause the processing device to perform operations.

Referring again briefly to FIG. 1, according to an exemplary embodiment, cabinet **102** also defines a mechanical compartment **170** at or near the bottom **106** of the cabinet **102** for receipt of a hermetically sealed cooling system **172**. In general, sealed cooling system **172** is configured for transporting heat from the inside of refrigerator appliance **100** to the outside (e.g., by executing a vapor-compression cycle or another suitable refrigeration cycle). As is generally understood by those of skill in the art, the hermetically sealed system **172** contains a working fluid, e.g., refrigerant, which flows between various heat exchangers of the sealed system **172** where the working fluid changes phases while transferring thermal energy.

In this regard, as understood by one having ordinary skill in the art, sealed system **172** may include a compressor, a condenser, an expansion device, and one or more evaporators connected in series by a fluid conduit that is charged with a refrigerant. Within sealed system **172**, refrigerant flows into the compressor, which operates to increase the pressure of the refrigerant. This compression of the refrigerant raises its temperature, which is lowered by passing the refrigerant through the condenser. Within the condenser, heat exchange with ambient air takes place so as to cool the refrigerant. A condenser fan may be used to pull air across the condenser, so as to provide forced convection for a more rapid and efficient heat exchange between the refrigerant within the condenser and the ambient air. Thus, as will be understood by those skilled in the art, increasing air flow across the condenser can, e.g., increase the efficiency of the condenser by improving cooling of the refrigerant contained therein.

An expansion device (e.g., an electronic expansion valve, capillary tube, or other restriction device) receives refrigerant from the condenser. From the expansion device, the refrigerant enters the evaporator. Upon exiting the expansion device and entering the evaporator, the refrigerant drops in pressure. Due to the pressure drop and/or phase change of the refrigerant, the evaporator is relatively cool. An evaporator fan is typically provided at each the evaporator, e.g., to force air across and around the at least one evaporator to transfer thermal energy from the air to the evaporator (and more particularly, to the working fluid or refrigerant therein).

In this manner, a flow of cooling air exits the evaporator and may be distributed to one or more of the chilled chambers **122** and/or **124**. Specifically, one or more ducts may extend between the mechanical compartment **170** and the chilled chambers **122** and/or **124** to provide fluid communication therebetween, e.g., to provide the chilled air from the hermetically sealed cooling system **172**, e.g., from an evaporator thereof, to one or more of the chilled chambers **122** and/or **124**.

The sealed system **172** described herein is provided by way of example only. Thus, it is within the scope of the

present subject matter for other configurations of the refrigeration system to be used as well. For example, according to alternative embodiments, sealed system **172** may include additional components, e.g., at least one additional evaporator, compressor, expansion device, and/or condenser. For example, refrigerator appliance **100** may have two or more split evaporators, e.g., one dedicated primarily to cooling fresh food chamber **122** and one dedicated primarily to cooling freezer chamber **124**. In addition, alternative plumbing configurations, valves, and flow regulators may be used to route refrigerant throughout sealed system **172**.

In some embodiments, refrigerator appliance **100** also includes one or more sensors that may be used to facilitate improved operation of refrigerator appliance **100**, such as described below. For example, in order to obtain temperature measurements within one or more chilled chambers **122**, **124** (or regions/zones within chilled chambers **122**, **124**), refrigerator appliance **100** may include a plurality of temperature sensors (not shown). Controller **156** may be communicatively coupled with the temperature sensors, may receive signals from these temperature sensors that correspond to the temperature of an atmosphere or air within their respective locations, and may implement responsive action, e.g., by directing more or less cooling air toward that region or chamber.

As used herein, "temperature sensor" or the equivalent is intended to refer to any suitable type of temperature measuring system or device positioned at any suitable location for measuring the desired temperature. Thus, for example, the temperature sensors may be any suitable type of temperature sensor, such as a thermistor, a thermocouple, a resistance temperature detector, etc. In addition, the temperature sensors may be positioned at any suitable location and may output a signal, such as a voltage, to a controller that is proportional to and/or indicative of the temperature of the air surrounding the temperature sensors. Although exemplary positioning of temperature sensors is described and illustrated herein, it should be appreciated that refrigerator appliance **100** may include any other suitable number, type, and position of temperature and/or other sensors according to alternative embodiments.

Referring still to FIG. 1, a schematic diagram of an external communication system **180** will be described according to an exemplary embodiment of the present subject matter. In general, external communication system **180** is configured for permitting interaction, data transfer, and other communications between refrigerator appliance **100** and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of refrigerator appliance **100**. In addition, it should be appreciated that external communication system **180** may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

For example, external communication system **180** permits controller **156** of refrigerator appliance **100** to communicate with a separate device external to refrigerator appliance **100**, referred to generally herein as an external device **182**. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network **184**. In general, external device **182** may be any suitable device separate from refrigerator appliance **100** that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard,

external device **182** may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home system, or another mobile or remote device.

In addition, a remote server **186** may be in communication with refrigerator appliance **100** and/or external device **182** through network **184**. In this regard, for example, remote server **186** may be a cloud-based server **186**, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device **182** may communicate with a remote server **186** over network **184**, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control refrigerator appliance **100**, etc. In addition, external device **182** and remote server **186** may communicate with refrigerator appliance **100** to communicate similar information.

In general, communication between refrigerator appliance **100**, external device **182**, remote server **186**, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device **182** may be in direct or indirect communication with refrigerator appliance **100** through any suitable wired or wireless communication connections or interfaces, such as network **184**. For example, network **184** may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

External communication system **180** is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system **180** provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

Referring now generally to FIGS. 2 and 3, refrigerator appliance **100** can further include an inventory management system **190** that is generally configured to monitor one or more chambers of refrigerator appliance **100** to monitor the addition and/or removal of inventory. More specifically, as described in more detail below, inventory management system **190** can include a plurality of sensors, cameras, or other detection devices that are used to monitor fresh food chamber **122** and/or freezer chamber **124** to detect objects **194** (e.g., food items, beverages) that are positioned in or removed from fresh food chamber **122** and/or freezer chamber **124**. In this regard, inventory management system **190** can use data from each of these devices to obtain a complete representation or knowledge of the identity, position, and/or other qualitative or quantitative characteristics of objects **194** within fresh food chamber **122** and/or freezer chamber **124**. Although inventory management system **190** is

described herein as monitoring fresh food chamber **122** for the detection of objects **194**, it should be appreciated that aspects of the present disclosure can be used to monitor objects or items in any other suitable appliance, chamber (e.g., freezer chamber **124**), etc.

Inventory management system **190** can include a camera assembly **192** coupled to refrigerator appliance **100** (e.g., to cabinet **102**) that is generally positioned and configured for obtaining images, video, and/or audio of refrigerator appliance **100** during operation. Specifically, according to the illustrated embodiment, camera assembly **192** includes one or more cameras that are mounted to cabinet **102**, to refrigerator doors **128**, or are otherwise positioned in view of fresh food chamber **122**. Although camera assembly **192** is described herein as being used to monitor fresh food chamber **122** of refrigerator appliance **100**, it should be appreciated that aspects of the present disclosure can be used to monitor any other suitable regions of any other suitable appliance, e.g., such as freezer chamber **124**. As best shown in FIGS. 2 and 3, camera assembly **192** is mounted to cabinet **102** at a front opening of fresh food chamber **122** and is oriented to have a field of view directed across the front opening and/or into fresh food chamber **122**.

Although camera assembly **192** is illustrated as having a single camera in FIGS. 2 and 3, it should be appreciated that camera assembly **192** can include a plurality of cameras positioned within and/or coupled (e.g., mounted) to cabinet **102**, wherein each of the plurality of cameras has a specified monitoring zone or monitoring range positioned around fresh food chamber **122**. In this regard, for example, the field of view of each camera assembly **192** can be limited to, directed to, or focused on a certain monitoring zone, monitoring range, or a specific area within fresh food chamber **122**. Specifically, cameras can be mounted to a sidewall of fresh food chamber **122** and can be spaced apart along the vertical direction V to cover different monitoring zones.

Notably, however, it can be desirable to position each camera assembly **192** proximate the front opening of fresh food chamber **122** and orient each camera assembly **192** such that the field of view of each camera assembly **192** is directed into fresh food chamber **122**. In this manner, privacy concerns related to obtaining images of the user of the refrigerator appliance **100** can be mitigated or avoided altogether. According to example embodiments, camera assembly **192** can be used to facilitate an inventory management process for refrigerator appliance **100**. As such, each camera assembly **192** can be positioned at an opening to fresh food chamber **122** to monitor objects **194** (e.g., food items, beverages) that are being added to or removed from fresh food chamber **122**.

According to still other embodiments, each camera assembly **192** can be oriented in any other suitable manner for monitoring any other suitable region within or around refrigerator appliance **100**. It should be appreciated that according to alternative embodiments, camera assembly **192** can include any suitable number, type, size, and configuration of camera(s) for obtaining images of any suitable areas or regions within or around refrigerator appliance **100**. In addition, it should be appreciated that each camera assembly **192** can include features for adjusting its field of view and/or orientation.

It should be appreciated that the images, video, and/or audio obtained by camera assembly **192** can vary in number, frequency, angle, resolution, detail, etc. in order to improve the clarity of the particular regions surrounding or within refrigerator appliance **100**. In addition, according to example embodiments, controller **156** can be configured to illuminate

the chilled chamber using one or more light sources prior to obtaining images. Notably, controller **156** of refrigerator appliance **100** (or any other suitable dedicated controller) can be communicatively coupled to camera assembly **192** and can be programmed or configured for analyzing the images obtained by camera assembly **192**, e.g., in order to identify items being added or removed from refrigerator appliance **100**, as described in detail below.

In general, controller **156** can be coupled (e.g., electrically, communicatively, operatively) to camera assembly **192** for analyzing one or more images, video, and/or audio obtained by camera assembly **192** to extract useful information regarding objects **194** located within fresh food chamber **122**. In this regard, for example, images, video, and/or audio obtained by camera assembly **192** can be used to extract a barcode, identify a product, monitor the motion of the product, or obtain other product information related to object **194**. Notably, this analysis can be performed locally (e.g., on controller **156**) or can be transmitted to a remote server (e.g., remote server **186** via external communication system **180**) for analysis. Such analysis is intended to facilitate inventory management, e.g., by identifying a food item being added to and/or removed from the chilled chamber.

Referring again to FIGS. **3** through **6**, a lighting assembly **200** that may be used with refrigerator appliance **100** will be described according to exemplary embodiments of the present subject matter. Specifically, as explained briefly above, lighting assembly **200** is generally intended to provide an improved user experience with refrigerator appliance **100**, e.g., by providing intuitive feedback to the user regarding the location of one or more objects (e.g., identified generally by reference numerals **194**) within refrigerator appliance **100**. In addition, lighting assembly **200** may be used to identify ingredients for a recipe, identify spoiled or expiring food, or to provide any other useful information to the user of refrigerator appliance **100**.

Specifically, as best shown in FIGS. **5** and **6**, lighting assembly **200** generally includes a plurality of light sources **204** positioned throughout fresh food chamber **122**. Notably, controller **156** may be configured for operating lighting assembly **200** in order to provide useful information to a consumer or user of refrigerator appliance **100**. In this regard, for example, it may be useful to a user of refrigerator appliance **100** to know the location of particular objects **194** within the chilled chamber. In this regard, according to exemplary embodiments, light sources **204** may be illuminated in those locations, e.g., to isolate and illuminate that particular zone to draw or focus a user's attention.

As used herein, the term "light sources" or the like may be used generally to refer to any suitable source of light for illuminating a refrigerator appliance **100** in any suitable manner. For example, light sources **204** may include any suitable number, type, position, and configuration of electrical light source(s), using any suitable light technology and illuminating in any suitable color. For example, according to the illustrated embodiment, light sources **204** includes one or more light emitting diodes (LEDs), which may each illuminate in a single color (e.g., white LEDs), or which may each illuminate in multiple colors (e.g., multi-color or RGB LEDs) depending on the control signal from controller **156**. However, it should be appreciated that according to alternative embodiments, light sources **204** may include any other suitable traditional light bulbs or sources, such as halogen bulbs, fluorescent bulbs, incandescent bulbs, glow bars, a fiber light source, etc. In addition, it should be appreciated that refrigerator appliance **100** may include

additional lighting, such as general chamber lighting that may illuminate the entire fresh food chamber **122** and/or freezer chamber **124**.

According to the illustrated embodiment, fresh food chamber **122** may generally comprise a plurality of storage locations (e.g., identified generally herein by reference numeral **210**). In this regard, storage locations **210** may be generally defined by bins **134**, shelves **136**, or other regions within fresh food chamber **122**. In this regard, referring briefly to FIG. **3**, storage locations **210** may generally be bounded by a liner of refrigerator appliance **100** and may be defined at least part by shelves **136** and/or a center mounting bracket **212** positioned on an illuminated rear wall **214** of fresh food chamber **122**. According to still other embodiments, each of these storage locations **210** may be subdivided into any suitable regions or zones. Accordingly, storage locations **210** generally refer to any location within fresh food chamber **122** where a food item or object **194** may be stored and which may be identified using lighting assembly **200**, as described in more detail below.

In this regard, referring still to FIG. **3**, lighting assembly **200** may include a plurality of lighting zones **220** that correspond to the plurality of storage locations **210**. In this regard, lighting zones **220** may generally be defined as the portions of illuminated rear wall **214** that correspond to a footprint of storage locations **210** when viewed from the front of refrigerator appliance **100**. As explained in more detail below, controller **156** may be in operative communication with lighting assembly **200** and may selectively illuminate lighting zones **220** to identify or draw a user's attention to a particular storage location **210**, e.g., a storage location **210** that contains an item or object **194** the user is wishing to access.

In addition to lighting zones **220** located on illuminated rear wall **214**, lighting assembly **200** may include other lighting zones **220** at other locations within refrigerator appliance **100**. For example, as shown in FIG. **3**, a top perimeter of each storage bin **134** may include a light strip or light sources **204** that define a bin lighting zone **220**. In addition, each of the storage shelves may include a front is that is illuminated as part of a lighting zone **220**, each drawer may have a lighting zone **220**, etc.

Although four exemplary lighting zones **220** are identified on the illuminated rear wall **214** of fresh food chamber **122** and one lighting zone **220** is identified on storage bin **134**, it should be appreciated that lighting assembly **200** may be split up into any other suitable number, size, and configuration of lighting zones **220** while remaining within the scope of the present subject matter. In addition, it should be appreciated that each of the lighting zones **220** may be configured to be independently illuminated and may be adjustable in at least one of color, intensity, flashing pattern, etc. Other variations to lighting assembly **200** are possible and within the scope of the present subject matter.

Referring now specifically to FIGS. **4** through **6**, light sources **204** of lighting assembly **200** may generally be oriented in a matrix of rows and columns. Lighting assembly **200** may further include a plurality of driver boards (not shown) for selectively operating one or more light sources **204**. Specifically, according to an example embodiment, each of the driver boards may be operably coupled to a subset of the matrix of light sources **204**. In this manner, controller **156** may operate the driver boards to selectively illuminate light sources **204**. In this manner, controller **156** may illuminate a single light source **204** or a group of light sources **204** as desirable to define one or more lighting zones **220**.

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As shown, illuminated rear wall **214** may generally include a backlight frame **230**, where the light sources **204** may be mounted directly to backlight frame **230**. In addition, a cover **232** may be positioned over backlight frame **230** and the plurality of light source **204** to generally form illuminated rear wall **214**. According to the illustrated embodiment, cover **232** may generally include a diffuser panel **234** that serves to diffuse light generated by light sources **204** and a transparent or semitransparent structural cover **236** positioned over diffuser panel **234**. In addition, cover **232** may include side trim pieces **238** that are generally configured for securing cover **232** to backlight frame **230** and to prevent light bleed out the edge of illuminated rear wall **214**. Other constructions of illuminated rear wall **214** are possible and within the scope of the present subject matter.

Notably, it may be desirable to prevent excessive light bleed between adjacent lighting zones **220**, e.g., to provide a more defined boundary of each lighting zone **220**. Accordingly, backlight frame **230** may generally include back panel **240** and a plurality of divider ribs **242** that extend from back panel **244** and cover **232**. In this regard, for example, divider ribs **242** may be positioned between the plurality of light sources **204** to define a particular illumination region for each particular light source **204**. For example, divider ribs **242** may extend along the horizontal direction between each row of light sources **204** and/or along the vertical direction V between each column of light sources **204**. In this manner, controller **156** may have discrete control of each light source **204** and may selectively illuminate light sources **204** to define any suitable lighting zone **220**.

According to the illustrated embodiment, divider ribs **242** may extend toward cover **232** and define a distal end **244**. As shown, a gap **246** may be defined between distal end **244** of divider ribs **242** and a front surface of backlight frame **230**. Notably, the size of gap **246** may be adjusted to regulate the amount of light bleed between lighting zones **220**. For example, it may be desirable to permit some light bleed to present a harsh dark line where divider ribs **242** contact cover **232**. By contrast, gap **246** may be designed such that there is a suitable amount of overlap to enable defined lighting zones **220**. For example, gap **246** may be between about 10% and 75%, between about 20% and 60%, between about 30% and 50%, or about 40% of a frame depth **248**.

Now that the construction and configuration of refrigerator appliance **100** and lighting assembly **200** have been presented according to an exemplary embodiment of the present subject matter, an exemplary method **300** for operating a lighting assembly in a refrigerator appliance is provided. Method **300** can be used to operate lighting assembly **200** or to operate any other lighting assembly. In this regard, for example, controller **156** may be configured for implementing method **300**. However, it should be appreciated that the exemplary method **300** is discussed herein only to describe exemplary aspects of the present subject matter and is not intended to be limiting.

As shown in FIG. 7, method **300** includes, at step **310**, receiving a request to identify an object stored in the chilled chamber. In this regard, continuing the example from above, a user of refrigerator appliance **100** may wish to access an object **194** that is positioned within fresh food chamber **122**. Accordingly, the user may communicate with controller **156** to request that object **194** be located and identified for the user to quickly and easily access the object **194**. According to example embodiments, the request to identify the object may include receiving a voice command from a user of refrigerator appliance **100**. For example, when a user opens refrigerator door **128**, a microphone (not shown) may listen

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for user prompts or requests. A user may state “where are my apples?” or may state “show me the eggs,” and the lighting zone **220** associated with those items may be illuminated.

According to still other embodiments, the request to identify the object of the chilled chamber may include detecting a user initiation of a recipe including the object. For example, a user may use a software application on remote device **182** to initiate a cooking operation that involves a recipe. Controller **156** may detect the initiation of the recipe and may automatically illuminate all ingredients needed for the recipe.

According to still other embodiments, the request to identify an object may be automated based on the expiration or spoilage of an object **194** within fresh food chamber **122**. For example, if the milk or eggs are approaching their expiration date, the corresponding lighting zone **220** may flash or indicate a warning color to prompt the user to remove such items. For example, controller **156** may implement an inventory management system for tracking system that has knowledge of when an item was added and its expiration date. Controller **156** may thereby identify food items that are approaching their expiration and may illuminate a target lighting zone **220** associated with these items.

Referring again to FIG. 7, step **320** may generally include determining that the object is located within a target location of the plurality of storage locations. In this regard, controller **156** may use an inventory management system **190** or an inventory tracking system to obtain the specific storage location **210** of the object **194** being requested. In this regard, as explained briefly above, refrigerator appliance **100** may include a camera assembly **192** may be generally used to track inventory as it passes into and/or out of fresh food chamber **122**. This location information may be used to identify a target storage location **210** and corresponding lighting zone **220**.

Step **330** may include identifying a target lighting zone of the plurality of lighting zone that corresponds to the target location of the object. Once the target lighting zone has been identified, step **340** may include illuminating the target lighting zone. It should be appreciated that the illumination of the target lighting zone may include changing the operation of light sources **204** to identify a location of object **194**. For example, light sources **204** not located within the target lighting zone **220** may maintain general chamber illumination while light sources **204** within the target lighting zone **220** may illuminate in a manner that differentiates that particular zone, such as supplying brighter light, a different color, a flashing light, etc.

Notably, method **300** may further include a calibration process for identifying the location and mapping of each light source **204** of lighting assembly **200**. In this manner, controller **156** may selectively illuminate each of light sources **204** and may monitor the illumination of light sources **204** using camera assembly **192**. In this manner, controller **156** may match the locations of each light source for later recall when implementing method **300**. In addition, camera assembly **192** may be used to identify a position of shelves **136**, bins **134**, or other storage features within refrigerator appliance **100**. In this manner, each of the plurality of lighting zones **220** may be adjusted based on the position of such shelves and bins.

FIG. 7 depicts an exemplary control method having steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the steps of any of the methods discussed herein can be adapted, rearranged, expanded, omitted, or modified in various ways

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without deviating from the scope of the present disclosure. Moreover, although aspects of the methods are explained using refrigerator appliance **100** and lighting assembly **200** as an example, it should be appreciated that these methods may be applied to the operation of any suitable appliance and/or lighting assembly.

As explained herein, aspects of the present subject matter are directed to a lighting system in a refrigerator that selectively lights certain areas to guide a consumer to a particular location (e.g., a particular shelf, one side of the refrigerator, a particular drawer or bin, etc.) within the refrigerator. The lighting system may be broken up into multiple smaller zones that individually illuminate with different colors and/or intensities. The zonal lighting may be used to highlight the location of food item that might be obstructed from the consumer's view and/or to aid in the location of the item.

The zonal back wall illumination system may include either an edge-lit style light or a traditional backlight where the optical axis of the LEDs projects from the back wall of a cavity directly through a diffusing element to illuminate the refrigerator cavity. A physical element may be positioned between each zone to prevent light bleeding from one zone to another unintentionally. The light sources may be RGB multi-color LEDs or individual monochromatic LEDs. The zones may be sized to match the height of the spacing between shelf locations and half of the cavity width. The individual zones may be much smaller and multiple zones may be illuminated together to give an appearance of a larger zone for a better spatial resolution if desired for the illumination.

In operation, consumers may select a food item to locate via the mobile software application and/or voice commands. An inventory management system may locate the food item within the refrigerator and illuminate the proper zone with a different color to bring attention to that area within the refrigerator. The notification color may be predetermined or customizable by the consumer. The inventory management system may give a different illuminated notification automatically when the consumer opens the refrigerator to indicate a food item that is expiring soon. The separate lighting system may also be applied underneath door bins to illuminate down into the bin to highlight food items on the doors.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A refrigerator appliance comprising:

- a cabinet;
- a chilled chamber defined within the cabinet, the chilled chamber comprising a plurality of storage locations;
- a lighting assembly comprising a plurality of lighting zones corresponding to the plurality of storage locations, the lighting assembly comprises:
  - a backlight frame comprising a back panel and a plurality of divider ribs extending from the back

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panel toward a cover and being positioned between the plurality of light sources to define the plurality of lighting zones;

a plurality of light sources mounted on the backlight frame; and

the cover positioned over the backlight frame and the plurality of light sources; and

a controller in operative communication with the lighting assembly, the controller being configured to:

receive a request to identify an object stored in the chilled chamber;

determine that the object is located within a target location of the plurality of storage locations;

identify a target lighting zone of the plurality of lighting zones that corresponds to the target location; and

illuminate the target lighting zone.

2. The refrigerator appliance of claim 1, wherein the request to identify the object stored in the chilled chamber comprises:

receiving a voice command from a user of the refrigerator appliance.

3. The refrigerator appliance of claim 1, wherein the request to identify the object stored in the chilled chamber comprises:

detecting a user initiation of a recipe including the object.

4. The refrigerator appliance of claim 1, wherein the controller is in operative communication with a remote device through an external network, and wherein the request to identify the object stored in the chilled chamber is provided through the remote device.

5. The refrigerator appliance of claim 1, wherein determining that the object is located within the target location of the plurality of storage locations comprises:

obtaining the target location using an inventory tracking system of the refrigerator appliance.

6. The refrigerator appliance of claim 1, wherein each of the plurality of lighting zones is configured to be independently illuminated.

7. The refrigerator appliance of claim 1, wherein the lighting assembly comprises a plurality of light sources that are adjustable in at least one of color or intensity.

8. The refrigerator appliance of claim 1, wherein the controller is further configured to:

identify a food item that is approaching expiration or spoilage; and

illuminating the target lighting zone containing the food item.

9. The refrigerator appliance of claim 1, wherein the lighting assembly comprises:

at least one of an illuminated rear wall, illuminated storage bins, or illuminated drawers.

10. The refrigerator appliance of claim 1, wherein the lighting assembly comprises:

a plurality of light sources oriented in a matrix of rows and columns; and

a plurality of driver boards, each of the plurality of driver boards being operatively coupled to a subset of the matrix of the plurality of light sources.

11. The refrigerator appliance of claim 1, wherein the plurality of divider ribs extend to a distal end, and wherein a gap is defined between the distal end of plurality of ribs and a front surface of the backlight frame.

12. The refrigerator appliance of claim 11, wherein the gap is between about 30% and 50% of a frame depth.

13. The refrigerator appliance of claim 1, further comprising a camera assembly coupled to the cabinet and operable to monitor the chilled chamber, the controller being

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in operative communication with the camera assembly and being configured to perform a calibration process, the calibration process comprising:

selectively illuminating a plurality of light sources of the lighting assembly; and

monitoring the illumination of the plurality of light sources using the camera assembly to map locations of the plurality of light sources as the plurality of light sources are selectively illuminated.

**14.** The refrigerator appliance of claim **1**, further comprising a camera assembly coupled to the cabinet and operable to monitor the chilled chamber, the controller being in operative communication with the camera assembly and being configured to:

identify a position of a shelf within the chilled chamber; and

adjust the plurality of lighting zones based on the position of the shelf.

**15.** A method of operating a lighting assembly of a refrigerator appliance, the refrigerator appliance comprising a chilled chamber defined within a cabinet, the chilled chamber comprising a plurality of storage locations, the lighting assembly comprising a plurality of lighting zones corresponding to the plurality of storage locations, and a camera assembly coupled to the cabinet and operable to monitor the chilled chamber, the method comprising:

performing a calibration process comprising:

selectively illuminating a plurality of light sources of the lighting assembly; and

monitoring the illumination of the plurality of light sources using the camera assembly to map locations of the plurality of light sources as the plurality of light sources are selectively illuminated;

receiving a request to identify an object stored in the chilled chamber;

determining that the object is located within a target location of the plurality of storage locations;

identifying a target lighting zone of the plurality of lighting zones that corresponds to the target location; and

illuminating the target lighting zone.

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**16.** The method of claim **15**, wherein the request to identify the object stored in the chilled chamber comprises: receiving a voice command from a user of the refrigerator appliance or detecting a user initiation of a recipe including the object.

**17.** The method of claim **15**, wherein each of the plurality of lighting zones is configured to be independently illuminated and is adjustable in at least one of color or intensity.

**18.** A refrigerator appliance comprising:

a cabinet;

a chilled chamber defined within the cabinet, the chilled chamber comprising a plurality of storage locations;

a lighting assembly comprising a plurality of lighting zones corresponding to the plurality of storage locations;

a camera assembly coupled to the cabinet and operable to monitor the chilled chamber; and

a controller in operative communication with the lighting assembly, the controller being configured to:

receive a request to identify an object stored in the chilled chamber;

determine that the object is located within a target location of the plurality of storage locations;

identify a target lighting zone of the plurality of lighting zones that corresponds to the target location;

illuminate the target lighting zone;

identify a position of a shelf within the chilled chamber; and

adjust the plurality of lighting zones based on the position of the shelf.

**19.** The refrigerator appliance of claim **18**, wherein the request to identify the object stored in the chilled chamber comprises:

receiving a voice command from a user of the refrigerator appliance.

**20.** The refrigerator appliance of claim **18**, wherein the request to identify the object stored in the chilled chamber comprises:

detecting a user initiation of a recipe including the object.

\* \* \* \* \*